

PROJECT REPORT

Luminous Beats: Music Reactive LED Controller

A Purely Analog, Transistor-Based Sound-to-Light System

Course: Sessional on Electronics-I (EE2110)

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1. Introduction

Sound-responsive lighting systems represent one of the most visually engaging applications of basic electronics. This project, titled Luminous Beats, demonstrates how sound energy from music or any audio source can be directly converted into a dynamic light display using fundamental analog components — no microcontrollers, no programming, and no digital processing.

The system is built around the HW-484 V0.2 sound sensor module, which integrates an electret condenser microphone, a bias resistor, internal signal conditioning circuitry, and analog output onto a single PCB. The HW-484 captures ambient sound and outputs a varying analog voltage (AO pin) directly proportional to sound intensity. This module replaces what would otherwise require a manually built preamplifier. The AO output feeds into a chain of five BC547 NPN transistors, each acting as an electronic switch. Five LEDs of different colors are each driven by one transistor.

The project demonstrates three core principles:

- (1) transduction (sound pressure → electrical signal via the microphone)
- (2) transistor switching (base current drives saturation/cutoff modes) and
- (3) analog signal conditioning

The end result is a circuit that pulses LEDs in real time with music rhythm and intensity. This project is particularly suited for a course on transistors and diodes, as it provides direct, observable evidence of transistor switching behavior driven by a real-world input signal.

2. Literature Review

2.1 Introduction to Sound Detection

Sound detection using electronic circuits has a long history. Electret condenser microphones, embedded in the HW-484 module, were first described by Sessler and West (1962) in the Journal of the Acoustical Society of America. Since then, sound detection has become fundamental to security systems, noise monitoring, biomedical equipment, and consumer electronics. The electret microphone — requiring only a bias resistor and 3-5V — is the most widely adopted acoustic transducer in educational electronics.

2.2 Evolution of Transistor Switching

Bipolar junction transistors (BJTs) as electronic switches are foundational to electronics engineering. Sedra and Smith (2020) in Microelectronic Circuits explain BJT operation in saturation and cutoff regions. The BC547 NPN transistor has an h_{FE} or β (current gain) of 110-800 and maximum collector current of 100 mA, sufficient for driving standard 5 mm LEDs requiring 10-20 mA.

2.3 Sensor Module Integration vs. Discrete Design

Sound-reactive LED circuits vary in complexity. Simple approaches use a single transistor with a biased microphone, working in loud environments but failing for soft sounds (microphone output too weak). Integrated sensor modules like the HW-484 incorporate internal amplification and signal conditioning, providing usable output without requiring a discrete preamplifier. This project initially aimed for a hand-built preamplifier but pivoted to the HW-484 when component availability became constrained. The decision illustrates an important principle: choosing the right integrated solution beats building everything from scratch when the goal is a working, robust system.

3. Components, Specifications, and Pinouts

3.1 Component List

Component	Specifications / Rating	Qty
HW-484 V0.2 Sound Sensor	Electret mic + internal conditioning, 3.3-5V, AO output	1
BC547 NPN Transistor	$\beta = 110-800$, $V_{CEO} = 45V$, $I_{C(max)} = 100mA$, TO-92	5
LM7805 Voltage Regulator	78xx series, 5V output, 1A max, TO-92	1
LED - Red/Green/Yellow/Blue	5mm, forward voltage 1.8-3.4V, 20mA each	5
Resistor - 47k Ω	1/4W, $\pm 5\%$, series resistor on AO line	1
Resistor - 220 Ω	1/4W, $\pm 5\%$, LED current-limiting	5
9V Battery	PP3 alkaline, 9V DC	1
Breadboard + Jumpers	400-point mini breadboard, male-to-male wires	1

3.2 Detailed Component Descriptions and Pinouts

3.2.1 HW-484 V0.2 Sound Sensor Module

The HW-484 is an integrated sound detection module that replaces a discrete preamplifier circuit. It combines four functional blocks on a single PCB: an electret condenser microphone, bias network, internal signal conditioning, and a potentiometer for sensitivity adjustment.

Electret Condenser Microphone: A permanently charged polymer diaphragm is positioned near a fixed backplate, forming a variable capacitor. When sound vibrates the diaphragm, capacitance changes, producing a small voltage variation (millivolt range). A built-in JFET transistor provides initial impedance transformation.

Internal Bias Resistor: A resistor on the module (typically 10k Ω) connects between VCC and the microphone output, supplying the DC bias current required by the JFET inside the microphone capsule. Without this bias, the JFET cannot function.

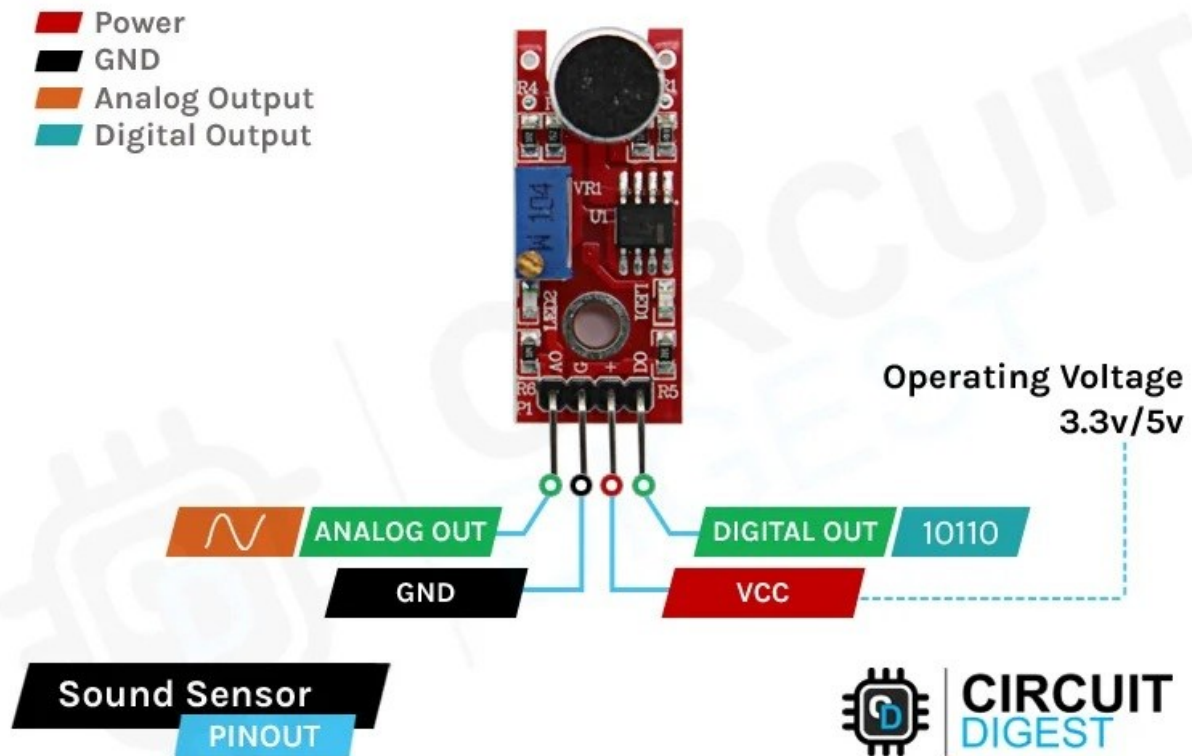
Analog Output (AO) Path: The microphone signal passes through a basic amplification stage and appears on the AO output pin as a DC-biased varying voltage. At silence, this sits at approximately 2-4V. As sound intensity increases, the voltage swings above and below this operating point, with magnitude proportional to sound amplitude. This output is AC (varying) superimposed on a DC bias, ready to drive a transistor base directly through a limiting resistor.

Trimmer Potentiometer: The blue multi-turn potentiometer sets the gain of the internal conditioning circuit. Rotating it clockwise increases sensitivity for quieter sounds; counter-clockwise reduces it to prevent false triggering in noisy environments.

Digital Output (DO) Path: The module also provides a digital output pin (DO), generated using an onboard voltage comparator circuit. The comparator continuously compares the microphone signal with a reference threshold set by the trimmer potentiometer.

- When the sound level exceeds the preset threshold, the DO pin changes state (HIGH or LOW depending on module design).
- When the sound level falls below the threshold, the output returns to its original state.

This digital output is mainly used for simple sound detection applications such as clap switches, alarms, or microcontroller interrupt inputs. However, in this project, the DO pin was not used because the objective was to preserve continuous analog sound variation rather than simple ON/OFF detection.



HW-484 Pin Diagram:

- VCC ——— Supply voltage input (3.3-5V from LM7805)
- GND ——— Ground reference
- AO ——— Analog Output (used in this circuit)
- DO ——— Digital Output (not used)

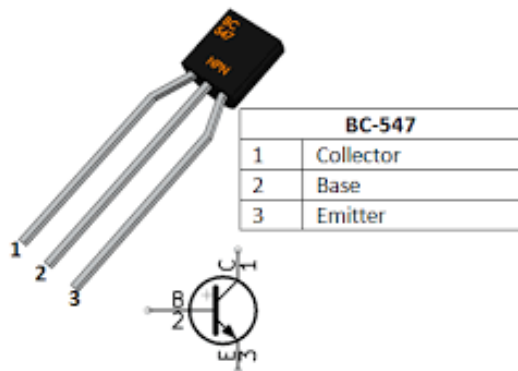
3.2.2 BC547 NPN Transistor (Common-Emitter Configuration)

The BC547 is a small-signal NPN transistor used for switching. In this circuit, all five BC547s operate as electronic switches in saturation/cutoff two-state mode, not as linear amplifiers.

Package: TO-92 plastic package with three leads. Viewed from the flat (marked) face with legs pointing downward, the pin order left to right is: Collector, Base, Emitter.

Operating Principle: When AO output rises above approximately 0.6V, sufficient base current flows through the 47kΩ resistor. This drives the transistor into saturation, creating a low-impedance path between collector and emitter ($V_{CE(sat)} \approx 0.2V$). The transistor acts as a closed switch. When AO voltage drops, base current falls and the transistor enters cutoff, behaving as an open switch with no collector current.

BC547 Pin Diagram (TO-92, flat face view):



Here:

C = Collector (to LED circuit)

B = Base (from 47kΩ resistor → AO pin)

E = Emitter (to 220Ω resistor → GND)

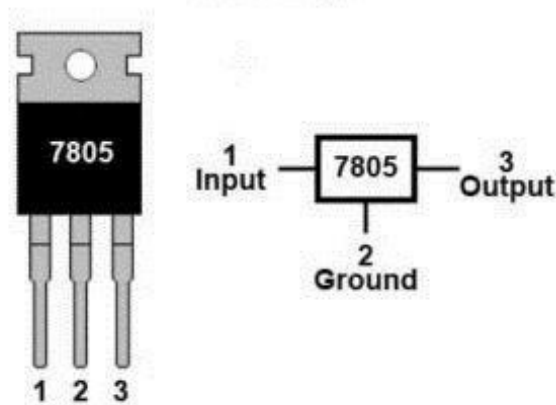
3.2.3 LM7805 Voltage Regulator (Linear Regulator)

The LM7805 is a three-terminal fixed-output linear voltage regulator from the 78xx series. It maintains constant 5V output regardless of input voltage variations or load changes, using an internal reference voltage, error amplifier, and pass transistor.

Why Needed: The project uses a 9V battery as main power. The HW-484 module requires 3.3-5V on its VCC pin. A direct 9V connection would overvoltage the internal circuitry and damage the module. The LM7805 drops 9V to 5V and supplies the HW-484, while the transistor-LED chain operates directly from 9V for maximum LED brightness.

Internal Operation: The LM7805 contains a precision 5V reference and an error amplifier that continuously compares the output voltage against this reference. A high-gain pass transistor adjusts its conductivity to keep the output at exactly 5V. With a 9V input, the excess voltage (9V - 5V = 4V) is dissipated as heat in the regulator body. Output current is limited to 1A, sufficient for the HW-484 (total draw < 100mA).

7805 Pinout



Here:

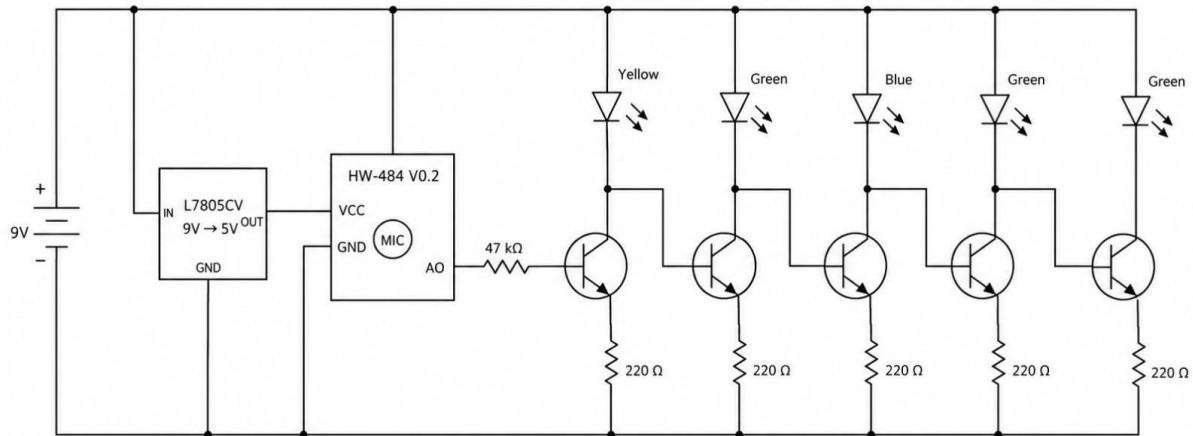
IN = Input (from 9V battery)

GND = Ground (common reference)

OUT = Output (+5V, to HW-484 V_{CC})

4. Project Details

4.2 Circuit Diagram



Sound responsive LED circuit

4.2 Circuit Architecture

The circuit operates through three functional stages: (1) Sound detection and amplification (HW-484 module), (2) Base current limiting (47kΩ resistor), and (3) LED switching (BC547 transistors and LEDs).

4.3 Working Process

Stage	Name	Components	Function
1	Sound Capture	HW-484 module	Detects sound, Amplifies it and outputs AO voltage
2	Base Limiting	47kΩ resistor	Protects HW-484; sets base current
3	LED Switching	5× BC547 + LEDs + 220Ω	Switches LEDs ON/OFF by sound

Stage 1 — Sound Capture (HW-484 Module): Sound waves strike the electret condenser microphone. The diaphragm vibrates, producing a tiny voltage variation. The internal bias resistor polarizes the microphone JFET buffer, and the internal signal conditioning circuit amplifies and conditions this into a DC-biased varying voltage on the AO pin. At silence, voltage sits at approximately 2-4V; it varies above/below this with sound intensity. The blue potentiometer allows sensitivity adjustment for the ambient environment.

Stage 2 — Base Current Limiting (47kΩ Resistor): The AO pin output connects through a 47kΩ series resistor to the junction of all five transistor base pins. This resistor limits base current: with approximately 5V on AO and 0.7V base-emitter threshold, $I_B \approx (5V - 0.7V) / 47k\Omega$

≈ 91 μA per transistor (divided among five in parallel). This current is sufficient to saturate each BC547 when AO is high, while keeping the HW-484 output current within safe limits.

Stage 3 — LED Switching (5× BC547 + 5× LEDs): The AO signal distributes to the base pins of five BC547 NPN transistors in parallel. Each operates in common-emitter configuration as a switch, controlling one LED in series with a 220Ω current-limiting resistor between +9V and the transistor collector. When AO voltage rises above 0.7V, sufficient base current drives each BC547 into saturation, switching all five LEDs ON simultaneously. When sound stops and AO drops, base current falls to zero, transistors enter cutoff, and LEDs extinguish. The 220Ω resistors protect LEDs by limiting collector current to approximately

$$\frac{(9V - V_{LED} - V_{CE(sat)})}{220\Omega} \approx 30 \text{ mA}.$$

4.3 Transistor Operation in Detail

Mode of Operation — Switching: All five BC547 transistors operate as digital switches in two-state operation: fully ON (saturation) or fully OFF (cutoff). They are not used as linear amplifiers. This is fundamental: an amplifier operates in the active region, faithfully reproducing and enlarging input signals. A switch operates in saturation/cutoff, regardless of exact input magnitude — it simply determines whether collector current flows.

Operating Regions — Cutoff (LED OFF): When base-emitter voltage $V_{BE} < 0.6V$ (no sound), there is no base current I_B . Therefore, collector current $I_C \approx 0$. The transistor is an open circuit between collector and emitter. $V_{CE} \approx 9V$ (approximately the full supply voltage).

Operating Regions — Saturation (LED ON): When sound is detected, AO voltage rises and drives V_{BE} above 0.7V through the 47kΩ resistor. Sufficient I_B flows to produce I_C that satisfies load requirements. In saturation, the transistor is effectively a closed switch with $V_{CE(sat)} \approx 0.2V$. Collector current is determined by external circuit $(9V - V_{LED} - V_{CE(sat)}) / 220\Omega$, not by $\beta \times I_B$, because the transistor is fully ON.

Configuration — Common Emitter: Each BC547 is wired in common-emitter (CE) configuration: input signal (AO) applied between base and emitter; output (LED + 220Ω) connected between collector and positive supply; emitter connected to limiting resistor then ground, which is common to both input and output loops. Common-emitter is the most widely used configuration for transistor switches.

Biasing Arrangement: This design uses no complex static DC biasing network. Transistors are intentionally biased at cutoff in the absence of signal. The only biasing element is the 47kΩ series resistor between AO and transistor bases, limiting base current when AO is high. This is base-resistor biasing — the simplest arrangement, appropriate for switching applications where transistors operate between cutoff and saturation, not in the linear active region.

5. Results and Discussion

Upon completing circuit assembly with the HW-484 V0.2 module, LM7805 regulator, five BC547 transistors in common-emitter configuration, five LEDs, and 220Ω resistors, the system was powered with a 9V battery and tested with music and voice at varying distances. The following observations were recorded:

Observation 1 — Real-time LED response: All five LEDs responded visually to sound within milliseconds, with no perceptible delay between a sound event (clap or bass note) and the LED flash. This confirmed that an all-analog signal path — from HW-484 AO through 47kΩ resistor to transistor bases — has inherently near-zero latency compared to microcontroller-based equivalents requiring ADC sampling and processing.

Observation 2 — All-or-none LED behavior: Because all five transistors share the same base drive signal from AO (through 47kΩ), all five LEDs switch ON and OFF together. This is

consistent with the switching design: once AO voltage exceeds 0.7V threshold, all transistors saturate simultaneously.

Observation 3 — Sensitivity adjustment: The blue potentiometer on the HW-484 proved essential for calibrating the circuit to the environment. In a quiet room, maximum sensitivity allowed response to whispered speech at approximately 30 cm. In noisy environments, reducing sensitivity prevented false triggering.

Observation 4 — Voltage regulator stability: The LM7805 provided stable 5V to the HW-484 throughout testing. Without the regulator, connecting 9V directly to the module VCC would overvoltage the internal circuitry and cause erratic behavior or damage. The regulator was confirmed by measuring module VCC with a multimeter: 4.98V.

Learning 1 — Transistor as a switch: The most significant practical insight was directly observing BC547 operating in its two switching states. At no sound, measuring VCE showed full supply voltage (open/cutoff). With sound detected, VCE dropped to approximately 0.2V (closed/saturation). This directly demonstrated the switching characteristic described in transistor datasheets.

Learning 2 — Role of the HW-484 module: The original circuit concept required a manually built preamplifier (BC547 in active region with biasing resistors and coupling capacitors) to amplify the weak microphone signal. By using the HW-484 module, this entire signal conditioning front-end is handled by the module internals. This simplified breadboard assembly and eliminated risks of manual preamplifier design errors. The decision taught an important lesson: understanding what the system needs (DC-blocked, conditioned AC signal) is more important than insisting on how to build it.

Learning 3 — Common-emitter configuration: Wiring BC547 in common-emitter configuration reinforced the pin ordering (Collector, Base, Emitter — flat face left to right) and current flow direction described in course theory. Reversing a transistor connection was the most common assembly error, immediately identifiable because the LED remained permanently ON or permanently OFF regardless of sound.

6. Conclusion

This project successfully designed, assembled, and tested a purely analog music-reactive LED controller using the HW-484 V0.2 integrated sound sensor module, LM7805 voltage regulator, and five BC547 NPN transistors in common-emitter switching configuration. Beginning with the fundamental physics of sound — pressure waves causing diaphragm displacement in the electret condenser microphone — the project traced the complete signal path: acoustic transduction inside the HW-484 module, internal amplification and conditioning, analog output voltage delivery through the 47kΩ base resistor, and finally transistor switching to drive the LEDs.

A key design decision was the substitution of a manually built discrete preamplifier circuit (originally specified in the project concept) with the HW-484 V0.2 integrated sensor module. The HW-484 contains an electret microphone with JFET buffer, bias resistor, and internal signal conditioning circuitry. This module performs the same signal amplification and conditioning function as a discrete preamplifier stage. The final implementation requires only five BC547 transistor switch stages, removing the complexity of discrete amplifier design while preserving the core educational objectives.

All five BC547 transistors operate in common-emitter configuration, switching between cutoff (LED OFF, no base current) and saturation (LED ON, $V_{CE} \approx 0.2V$) in response to the AO output of the sensor module. Base-resistor biasing through the 47kΩ resistor keeps transistors in cutoff at silence and drives them into saturation when sound is present. The LM7805 linear voltage regulator provides a stable 5V supply to the HW-484 module from the 9V battery, protecting the module from overvoltage.

The circuit achieved its primary objective: LEDs responded in real time to music rhythm and intensity with no programming, no microcontroller, and no digital components. The project provided hands-on understanding of transducer operation, BJT saturation and cutoff switching, common-emitter configuration, base-resistor biasing, linear voltage regulation, and LED current limiting — concepts directly aligned with the course objectives of Basics of Transistors and Diodes.

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